Metals, acids, salts and chemical tests

30 questions: 10 multiple choice, 20 structured.

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1 · Part 2

Paper 2 only Extraction and uses of metals [1]

Which metal is most likely to be found in the Earth's crust as the uncombined element?

A aluminium

B iron

C gold

D zinc

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2 · Part 2

Paper 2 only Extraction and uses of metals [1]

Why is an alloy usually harder than the pure metal it is made from?

A Atoms of a different size disrupt the regular layers, so the layers cannot slide over each other as easily.

B The atoms in an alloy are held together by covalent bonds, which are stronger than metallic bonds.

C An alloy has fewer delocalised electrons, so the electrons can move through the structure more easily.

D The atoms in an alloy are further apart, so the forces between them are weaker.

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3 · Part 2

Paper 2 only

Extraction and uses of metals

[7]

The table shows part of the reactivity series, with carbon included. The most reactive element is at the top.

Element
potassium
sodium
calcium
magnesium
aluminium
carbon
zinc
iron
copper
gold

(a) Iron is extracted by heating iron(III) oxide with carbon. Explain, using the reactivity series, why carbon can be used to extract iron from iron(III) oxide. [2]

(b) During the extraction of iron, carbon monoxide reacts with iron(III) oxide. State which substance is reduced in this reaction. Give a reason for your answer.

$Fe_2O_3(s) + 3CO(g) \rightarrow 2Fe(l) + 3CO_2(g)$ [2]

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(c) Aluminium is extracted from aluminium oxide by electrolysis. Explain why aluminium cannot be extracted by heating aluminium oxide with carbon. [2]

(d) Give a reason why gold does not need to be extracted from a compound. [1]

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4 · Part 2

Paper 2 only Extraction and uses of metals [8]

Titanium is a strong metal with a low density. It is extracted from titanium(IV) oxide, TiO_2, in a batch process. The information describes the four stages of the process.

Stage 1: Titanium(IV) oxide is heated with carbon and chlorine to make titanium(IV) chloride, $TiCl_4$.

Stage 2: The titanium(IV) chloride is purified by distillation.

Stage 3: The titanium(IV) chloride is heated to about 900 °C with molten magnesium in a sealed steel container filled with argon. Titanium and magnesium chloride are formed. This stage takes several days.

Stage 4: The magnesium chloride is electrolysed to produce magnesium and chlorine. These are used again in the process.

(a) Complete the balanced equation for the reaction in Stage 3.

$TiCl_4$ + ...Mg → Ti + ...$MgCl_2$ [1]

(b) Deduce what Stage 3 shows about the reactivity of titanium compared with the reactivity of magnesium. Give a reason for your answer. [2]

(c) Suggest why Stage 3 is carried out in an atmosphere of argon rather than in air. [2]

(d) Titanium is much more expensive than iron. Iron is made in a continuous process by heating its oxide with carbon. Comment on why titanium is expensive to produce. Use the information about the process. [3]

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5 · Part 2

Paper 2 only Extraction and uses of metals [6]

Aluminium, copper, iron and steel have many uses because of their properties.

(a) The table shows four uses of metals. Complete the table by giving one property of each metal that makes it suitable for the use shown.

Use	Metal	Property
overhead power cables	aluminium	
electrical wiring in houses	copper	
cutlery and kitchen sinks	stainless steel	
drill bits and cutting tools	high-carbon steel	

[4]

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(b) Mild steel (low-carbon steel) is used to make car body panels. Explain why mild steel is more suitable than high-carbon steel for this use. [2]

6 · Part 2

Paper 2 only

Extraction and uses of metals

[6]

Brass is made from copper and zinc. Samples of brass containing different percentages of zinc were tested for hardness. A higher hardness number means a harder metal.

Percentage of zinc by mass	Hardness number
0	50
10	62
20	75
30	86
40	98

(a) Brass is an alloy. State what is meant by an alloy. [1]

(b) Describe the trend shown by the data. [1]

(c) Explain why brass is harder than pure copper. Refer to the arrangement of the particles in your answer. [3]

(d) Estimate the hardness number of brass that contains 25% zinc by mass. [1]

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7 · Part 2

Paper 1 & 2 | Acids, alkalis and titrations | [1]

Universal indicator is added to a solution. The colour of the indicator shows that the solution has a pH of 9.

How is this solution classified?

A strongly acidic

B neutral

C strongly alkaline

D weakly alkaline

8 · Part 2

Paper 1 & 2 | Acids, alkalis and titrations | [1]

Which row shows the colour of each indicator in sodium hydroxide solution?

	Litmus	Phenolphthalein	Methyl orange
A	red	colourless	red
B	blue	pink	yellow
C	blue	colourless	yellow
D	red	pink	red

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9 · Part 2

Paper 1 & 2 | Acids, alkalis and titrations | [6]

Hana uses universal indicator to find the pH of five solutions, J, K, L, M and N. The table shows her results.

Solution	pH
J	1
K	13
L	7
M	5
N	9

(a) Describe how Hana could use universal indicator paper to find the pH of solution J. [2]

(b) Identify the solution that is strongly alkaline. [1]

(c) Identify the solution that is weakly acidic. [1]

(d) State the formula of the ion that acids in aqueous solution are a source of. [1]

(e) State the formula of the ion that alkalis in aqueous solution are a source of. [1]

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10 · Part 2

Paper 1 & 2 | Acids, alkalis and titrations | [6]

Ali puts 20 cm^3 of dilute hydrochloric acid in a beaker. He adds sodium hydroxide solution, 4 cm^3 at a time, and measures the pH of the mixture with a pH probe after each addition.

Volume of sodium hydroxide solution added in cm^3	pH
0	1.0
4	1.1
8	1.3
12	1.6
16	2.2
20	11.9
24	12.4

(a) Describe how the pH of the mixture changes as the sodium hydroxide solution is added. [2]

(b) Identify the two volumes of sodium hydroxide solution between which the mixture became neutral. [1]

(c) Explain, in terms of ions, why the pH of the mixture increases as the sodium hydroxide solution is added. [2]

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(d) Suggest how Ali could find the volume of sodium hydroxide solution needed to make the mixture neutral more precisely. [1]

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11 · Part 2

Paper 2 only

Acids, alkalis and titrations

[7]

Zara wants to find the volume of dilute hydrochloric acid that exactly neutralises 25.0 cm^3 of sodium hydroxide solution.

She has this apparatus: a burette and clamp stand, a small funnel, a 25.0 cm^3 pipette with a pipette filler, a conical flask, a white tile and phenolphthalein indicator.

(a) Describe how Zara should carry out the titration. [5]

(b) Give a reason why Zara places the conical flask on a white tile. [1]

(c) Give a reason why Zara uses a pipette rather than a measuring cylinder to measure the sodium hydroxide solution. [1]

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12 · Part 2

Paper 2 only

Acids, alkalis and titrations

[8]

Leon titrates 25.0 cm^3 portions of potassium hydroxide solution with dilute sulfuric acid of concentration 0.100 mol/dm^3. He uses methyl orange indicator.

$2KOH(aq) + H_2SO_4(aq) \rightarrow K_2SO_4(aq) + 2H_2O(l)$

(a) The table shows Leon's burette readings. Complete the table by calculating the volume of acid added in each titration.

	Rough	**Titration 1**	**Titration 2**	**Titration 3**
final burette reading in cm^3	23.40	22.95	45.80	23.75
initial burette reading in cm^3	0.00	0.50	22.95	1.20
volume of acid added in cm^3				

[2]

(b) Concordant titres differ from each other by 0.10 cm^3 or less. Leon uses only his concordant titres to calculate the mean titre. Calculate the mean titre. [2]

(c) Calculate the concentration of the potassium hydroxide solution in mol/dm^3. [3]

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(d) Give a reason why Leon does not use the rough titre when he calculates the mean titre. [1]

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13 · Part 2

Paper 1 & 2 Acids, bases and salt preparations [1]

Which compound is insoluble in water?

A ammonium chloride

B calcium nitrate

C barium sulfate

D potassium carbonate

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14 · Part 2

Paper 1 & 2 Acids, bases and salt preparations [1]

Ammonia gas reacts with hydrogen chloride gas to form solid ammonium chloride.

$NH_3(g) + HCl(g) \rightarrow NH_4Cl(s)$

Which statement describes this reaction?

A Ammonia acts as an acid because it donates a proton.

B Hydrogen chloride acts as an acid because it donates a proton.

C Hydrogen chloride acts as a base because it accepts a proton.

D Ammonia acts as an acid because it accepts a proton.

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15 · Part 2

Paper 1 & 2 | Acids, bases and salt preparations | [1]

Which pair of substances reacts to form zinc sulfate, water and carbon dioxide?

A zinc and dilute sulfuric acid

B zinc carbonate and dilute hydrochloric acid

C zinc oxide and dilute sulfuric acid

D zinc carbonate and dilute sulfuric acid

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16 · Part 2

Paper 1 & 2 | Acids, bases and salt preparations | [7]

Dilute acids react with metals, bases and metal carbonates to form salts.

(a) The table shows four reactions. Complete the table by giving the name of the salt formed in each reaction.

Reactants	Name of salt formed
magnesium and dilute hydrochloric acid	
copper(II) oxide and dilute sulfuric acid	
calcium carbonate and dilute nitric acid	
potassium hydroxide and dilute sulfuric acid	

[3]

(b) State one observation that is made when calcium carbonate is added to dilute nitric acid. [1]

(c) Complete the balanced equation, including state symbols, for the reaction between magnesium and dilute hydrochloric acid.

Mg(s) + ...HCl(aq) → ..... [3]

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17 · Part 2

Paper 1 & 2 | Acids, bases and salt preparations | [6]

Bases react with acids to form salts. Some bases are also alkalis.

(a) State what is meant by an alkali. [1]

(b) The list shows five bases. Identify the two bases in the list that are alkalis.

copper(II) hydroxide, sodium hydroxide, iron(III) hydroxide, potassium hydroxide, magnesium hydroxide [2]

(c) When ammonia reacts with an acid, ammonium ions, NH_4^+, are formed. Explain, in terms of proton transfer, why ammonia acts as a base in this reaction. [2]

(d) Name the salt formed when ammonia reacts with dilute sulfuric acid. [1]

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18 · Part 2

Paper 1 & 2 | Acids, bases and salt preparations | [7]

Amir prepares crystals of hydrated copper(II) sulfate, $CuSO_4 \cdot 5H_2O$, from copper(II) oxide and dilute sulfuric acid.

He has a beaker, a measuring cylinder, a Bunsen burner, a tripod and gauze, a spatula, a glass rod, a filter funnel with filter paper, a conical flask and an evaporating basin.

Step 1: He warms 50 cm^3 of dilute sulfuric acid in the beaker.

Step 2: He adds copper(II) oxide powder, one spatula at a time, stirring after each addition, until the copper(II) oxide is in excess.

(a) Complete the equation for the reaction.

$CuO(s) + H_2SO_4(aq) \rightarrow$ ..... [1]

(b) Give a reason why Amir warms the acid in Step 1. [1]

(c) Give a reason why Amir adds the copper(II) oxide until it is in excess. [1]

(d) Describe how Amir should obtain pure, dry crystals of hydrated copper(II) sulfate from the mixture at the end of Step 2. [4]

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19 · Part 2

Paper 1 & 2 | Acids, bases and salt preparations | [6]

Priya wants to make a solution of zinc chloride. She could add zinc, zinc oxide or zinc carbonate to dilute hydrochloric acid.

(a) Identify which of these three substances reacts with dilute hydrochloric acid to form only zinc chloride and water. [1]

(b) Priya decides to use zinc carbonate. Complete the balanced equation, including state symbols, for the reaction.

$ZnCO_3(s) + 2HCl(aq) \rightarrow$ ..... [2]

(c) State how Priya can tell that the zinc carbonate is in excess. [1]

(d) Priya's friend wants to make copper(II) chloride by adding copper metal to dilute hydrochloric acid.

(i) Give a reason why this method will not work. [1]

(ii) Suggest a substance that the friend could add to dilute hydrochloric acid instead of copper to make copper(II) chloride. [1]

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20 · Part 2

Paper 2 only Acids, bases and salt preparations [7]

Ben wants to make a pure, dry sample of sodium sulfate crystals from sodium hydroxide solution and dilute sulfuric acid.

$2NaOH(aq) + H_2SO_4(aq) \rightarrow Na_2SO_4(aq) + 2H_2O(l)$

He has a burette and clamp stand, a 25.0 cm^3 pipette with a pipette filler, a conical flask, methyl orange indicator, an evaporating basin, a Bunsen burner, a tripod and gauze, a filter funnel and filter paper.

(a) When a salt is made from an insoluble base, an excess of the base is added to the acid and the excess is removed by filtering. Explain why Ben cannot use this method with sodium hydroxide. [2]

(b) Describe how Ben should use the apparatus to make pure, dry crystals of sodium sulfate. [5]

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21 · Part 2

Paper 2 only

Acids, bases and salt preparations

[7]

Jamal makes lead(II) sulfate by mixing lead(II) nitrate solution with sodium sulfate solution in a beaker.

He also has a glass rod, a filter funnel with filter paper, a conical flask, a wash bottle of distilled water and an oven.

(a) State what Jamal sees when he mixes the two solutions. [1]

(b) Complete the balanced equation, including state symbols, for the reaction.

$Pb(NO_3)_2(aq) + Na_2SO_4(aq) \rightarrow$ ..... [2]

(c) Describe how Jamal should obtain a pure, dry sample of lead(II) sulfate from the mixture. [4]

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22 · Part 2

Paper 2 only

Acids, bases and salt preparations

[7]

Ella makes the insoluble salt barium carbonate, $BaCO_3$, by mixing barium chloride solution with sodium carbonate solution.

$BaCl_2(aq) + Na_2CO_3(aq) \rightarrow BaCO_3(s) + 2NaCl(aq)$

The barium chloride solution contains 0.0120 mol of $BaCl_2$. Ella adds an excess of sodium carbonate solution.

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(a) Ella's friend suggests using calcium carbonate instead of sodium carbonate. Give a reason why calcium carbonate cannot be used in this method. [1]

(b) Calculate the maximum mass of barium carbonate that can be formed. Use the Periodic Table. [3]

(c) Ella obtains 2.05 g of dry barium carbonate. Calculate the percentage yield. [2]

(d) Suggest one reason why the percentage yield is less than 100%. [1]

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23 · Part 2

Paper 1 & 2 | Acids, bases and salt preparations | [6]

Tariq uses the general rules for the solubility of ionic compounds in water.

(a) Complete the table to show whether each compound is soluble or insoluble in water.

Compound	Soluble or insoluble
potassium carbonate	
zinc nitrate	
silver chloride	
calcium sulfate	
magnesium hydroxide	
ammonium carbonate	

[3]

(b) Tariq adds dilute sulfuric acid to lumps of calcium carbonate. At first the mixture fizzes, but the fizzing soon stops even though some acid and some calcium carbonate are left.
Suggest why the reaction stops. [2]

(c) Tariq repeats the experiment using dilute hydrochloric acid instead of dilute sulfuric acid.
This time the reaction continues until all the calcium carbonate has reacted. Explain why. [1]

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24 · Part 2 Paper 1 & 2 Chemical tests [1]

Dilute nitric acid and then silver nitrate solution are added to a solution of a compound. A cream precipitate forms.

Which ion is present in the compound?

A Br^-

B Cl^-

C I^-

D SO_4^{2-}

25 · Part 2 Paper 1 & 2 Chemical tests [1]

Sodium hydroxide solution is added to a solution of a compound. A green precipitate forms.

Which ion is present in the compound?

A Cu^{2+}

B Fe^{3+}

C Fe^{2+}

D NH_4^+

26 · Part 2 Paper 1 & 2 Chemical tests [1]

Which result shows that a colourless liquid is pure water?

A It turns anhydrous copper(II) sulfate from white to blue.

B It has a pH of 7.

C It freezes over a range of temperatures below 0 °C.

D It boils at exactly 100 °C at normal atmospheric pressure.

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27 · Part 2

Paper 1 & 2 | Chemical tests | [6]

Gases can be identified by simple tests.

(a) The table shows three gases. Complete the table by describing the test for each gas and its result.

Gas	Test	Result
oxygen		
hydrogen		
chlorine		

[3]

(b) Kofi adds dilute hydrochloric acid to a white solid. The mixture fizzes. Describe how Kofi could show that the gas given off is carbon dioxide. Give the result. [2]

(c) State what the result in (b) shows about the anion in the white solid. [1]

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28 · Part 2

Paper 1 & 2 | Chemical tests | [6]

Flame tests are used to identify some metal ions in compounds.

(a) Describe how to carry out a flame test on a solid sample. [3]

(b) The table shows the flame colours given by four samples. Complete the table by identifying the metal ion in each sample.

Sample	Flame colour	Metal ion
1	red	
2	lilac	
3	orange-red	
4	blue-green	

[3]

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29 · Part 2

Paper 1 & 2 | Chemical tests | [7]

Sofia has three solids, X, Y and Z. Each solid is an ionic compound that contains one type of cation and one type of anion. She dissolves each solid in water and tests the solutions. The table shows her results.

Test	X	Y	Z
add sodium hydroxide solution	blue precipitate	no precipitate	orange-brown precipitate
add dilute nitric acid, then silver nitrate solution	white precipitate	yellow precipitate	cream precipitate

In a flame test, solid Y gives a yellow flame.

(a) Identify X, Y and Z. Give the name of each compound. [3]

(b) Give the formula of the blue precipitate formed when sodium hydroxide solution is added to the solution of X. [1]

(c) Sofia's friend says that Z might contain iron(II) ions. Explain why Sofia's results show that this is wrong. [2]

(d) Give a reason why dilute nitric acid is added before the silver nitrate solution. [1]

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30 · Part 2

Paper 1 & 2 | Chemical tests | [7]

Lena has a fertiliser that she thinks is ammonium sulfate, $(NH_4)_2SO_4$. She dissolves some of the fertiliser in water.

(a) Describe a test to show that the solution contains ammonium ions. Give the result. [3]

(b) Describe a test to show that the solution contains sulfate ions. Give the result. [2]

(c) Lena then tests a solution of a second fertiliser, potassium nitrate, KNO_3.

(i) Give a reason why the test in (a) gives a negative result for potassium nitrate. [1]

(ii) Lena's teacher says that adding sodium hydroxide solution cannot be used to identify the potassium ions in potassium nitrate. Suggest why. [1]

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